

Per-domain compass heads

top-4 heads per bias domain, across four model families

Source: `helix_usage_validated/per_domain_alpha_{gpt2,phi3,gemma,llama}.json`. For every (model, domain) we list the top-4 passing heads used by the routed ensemble, ordered by calibrated contribution. ✓ = head passed the SNR gate at the calibrated α ; ✗ = retained fallback (SNR under target). α is the per-domain scale that hits the target SNR.

Model	Gender	Race	Profession	Religion
GPT-2	L9 H7 ✓, L11 H1 ✓ L8 H8 ✗, L7 H5 ✗ $\alpha=1.89$	L9 H7 ✓, L10 H1 ✓ L11 H1 ✓, L8 H10 ✗ $\alpha=1.80$	L9 H7 ✓, L10 H9 ✓ L11 H1 ✓, L7 H6 ✓ $\alpha=1.37$	L9 H2 ✓, L7 H5 ✗ $\alpha=1.87$
Phi-3-mini	L24 H14 ✓, L28 H26 ✓ L30 H4 ✓, L26 H22 ✓ $\alpha=6.38$	L24 H14 ✓, L27 H6 ✓ L28 H26 ✗, L26 H24 ✓ $\alpha=6.44$	L24 H14 ✓, L28 H1 ✓ L26 H24 ✓, L27 H16 ✓ $\alpha=4.75$	L28 H26 ✓, L24 H14 ✓ L29 H9 ✓, L30 H6 ✗ $\alpha=6.67$
Gemma-2-2B	L16 H2 ✓, L18 H2 ✗ L20 H3 ✗, L22 H1 ✗ $\alpha=55.26$	L20 H2 ✓, L24 H5 ✓ L25 H2 ✓, L18 H2 ✗ $\alpha=81.36$	L20 H5 ✓, L21 H4 ✓ L16 H2 ✓, L19 H4 ✗ $\alpha=29.73$	L14 H3 ✗, L14 H2 ✗ L16 H4 ✗, L7 H0 ✗ $\alpha=31.19$
Llama-3.2-3B	L22 H12 ✓, L22 H14 ✓ L26 H23 ✓, L23 H20 ✓ $\alpha=6.08$	L22 H12 ✓, L22 H14 ✓ L26 H21 ✓, L26 H23 ✓ $\alpha=5.34$	L22 H12 ✓, L22 H14 ✓ L26 H14 ✓, L20 H13 ✓ $\alpha=5.17$	L22 H12 ✓, L26 H20 ✗ L23 H14 ✗, L22 H13 ✗ $\alpha=6.99$

Reading guide. Each cell lists the 4 heads (layer-head) the routed ensemble uses for that (model, domain) pair, in rank order. The bottom row per model is the scalar α the per-domain calibrator selected to hit the target SNR (Gemma alphas are large because its residual-stream norms are substantially bigger than the other models'). Heads that recur across domains (e.g. GPT-2 L9 H7; Phi-3 L24 H14; Llama L22 H12, L22 H14) are the *multi-domain* heads; heads that appear in a single column are *domain-specific*. This is the source of the routing analysis in `head_domain_routing.md`: GPT-2 and Gemma are *specialized* (spec index ≥ 0.80), Phi-3 and Llama are *mixed* (spec index ≈ 0.50 , which is why both per-domain routing and per-domain α help those two).

Note on Gemma religion. None of the four top heads Gemma routes for religion pass the SNR gate — the fallback is retained with a high α . This is consistent with Gemma’s specialization profile (single dominant gender head at L21 H4) and explains the weaker StereoSet religion effect observed for Gemma.